VECTOR-SHIELD: SEMESTER 7 MASTER COMPLETION REPORT & SEMESTER 8 STRATEGIC ROADMAP

A Comprehensive Technical Audit, Gap Analysis, Publication Strategy, and Enterprise Productization Blueprint

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Supervisor & Guide:	**Dr. Lekshmi C. R.**, Assistant Professor, Department of Artificial Intelligence
Live Prototype:	https://vectorshield.streamlit.app/ | GitHub: https://github.com/VigneshVS2005/Vectorshield
Academic Status:	Semester 7 Final Evaluation Completed | Transitioning to Semester 8 Charter (Nov 2026)

1. Foundational Concept: What Our Project is All About

The Problem: Enterprise AI agents are rapidly adopting Autonomous Retrieval-Augmented Generation (RAG) to execute critical operations—such as bank wire transfers, database modifications, and cloud API management. However, these systems are fatally vulnerable to **Indirect Prompt Injections (IPI)** and **Vector Database Poisoning**. Attackers conceal imperative natural language commands within external documents (PDF vendor invoices, regulatory updates). When retrieved by semantic search, these instructions hijack the LLM control flow, causing unauthorized actions.

Why Existing Tools Fail: State-of-the-art defenses (Meta's Llama Guard 3, NeMo Guardrails) inspect text *after* generation using secondary LLMs. This incurs **800–2,500 ms of latency**, massive continuous token costs, and aborts client sessions with destructive hard-crash errors.

The Vector-Shield Solution: Vector-Shield introduces an in-flight, dual-tier neuro-symbolic firewall directly in high-dimensional vector space ($\blacksquare^{384} \rightarrow \blacksquare^{16}$) *before* prompt synthesis. In Stage 1, non-linear scaled LOF catches obvious outliers in **1.82 ms with zero token cost**. In Stage 2, borderline ambiguous payloads are classified by a fine-tuned **149M ModernBERT** model in 12.2 ms. An autonomous **LangGraph self-healing loop** dynamically blacklists poison IDs and re-retrieves clean context, while automated **SIEM webhooks** notify enterprise Security Operations Centers (SOC).

2. Semester 7 Completed Accomplishments (Module-by-Module Audit)

Module	Subsystem / Artifact	Key Technical Engineering Highlights	Empirical Status
M1	Mathematical Topographic Metric	LOF scaled by centroid drift (d_cos/τ)^1.2 and 16-D adaptive PCA projection fitted on clean vectors.	**100% Outlier Recall**
M2	Ground-Truth Dataset Scaler	175 authentic samples: 100 clean enterprise policies (SWIFT MT103, SOX, GDPR) + 75 Garak/HF injection payloads.	**175 Scaled Samples**
M3	Stage 1 Symbolic Interceptor	Implemented in src/lof_interceptor.py; 384-D MiniLM encoding, sub-2ms PCA density filter.	**1.82 ms Latency**
M4	Stage 2 ModernBERT Classifier	Fine-tuned answerdotai/ModernBERT-base (149M params, 8,192 window, FlashAttention-2) on NVIDIA A100 GPU (FP16 AMP).	**100.00% Holdout F1**
M5	LangGraph Self-Healing Loop	Cyclic DAG (retrieve → intercept → heal → respond); dynamic blacklist rewrites; zero-crash execution.	**Zero Crash Execution**
M6	Enterprise SIEM Telemetry	Automated JSON alert dispatching to Splunk/Sentinel/HTTPBin; verified HTTP 200 OK telemetry.	**HTTP 200 OK Delivery**
M7	Production SaaS Dashboard	Deployed on Streamlit Cloud with 6 operational tabs, 3D PCA vector space, and multi-chunk PDF ingestion.	**Live Public SaaS**
M8	B.Tech Thesis LaTeX Draft	7 full chapters, IEEE/Amrita thesis formatting, empirical manifold diagrams, and test suites.	**7 Full Chapters**

3. Semester 7 Gap Analysis & Honest Self-Critique (What Was Constrained)

To elevate this project to publishable tier and commercial viability in Semester 8, Team 14 conducted an honest, objective audit of the constraints maintained during Semester 7:

Gap ID	Identified Constraint in Sem 7	Root Cause / Technical Tradeoff	Semester 8 Remedy & Mitigation Plan
G1	**Dataset Scale:** 175 Curated Samples	Prioritized manual validation of authentic banking/regulatory texts for initial feasibility.	Scale to **10,000+ samples** across 4 enterprise domains (Banking, Healthcare, Legal, Code).
G2	**Text Modality Only:** 384-D Vectors	Focused on textual prompt injection vulnerabilities first.	Integrate **CLIP / SigLIP** multi-modal embeddings for visual prompt injections in scanned PDFs/charts.
G3	**In-Process Python Runtime**	Rapid prototyping in Python/Streamlit rather than compiled microservices.	Build an out-of-process **C++/Rust Envoy Sidecar** (<500µs latency) for Kubernetes pods.
G4	**Static Empirical Centroid (c)**	Offline fitting on 100 verified clean corporate documents.	Implement **Streaming Online Centroid Updates** (EMA) to adapt dynamically as documents evolve.
G5	**Absence of White-Box Attacks**	Tested empirical red-team sets (Garak) rather than direct gradient optimizers.	Stress-test against **Greedy Coordinate Gradient (GCG)** and HotFlip token optimization.
G6	**Single-Agent RAG Scope**	Evaluated on single-agent retrieve-and-generate workflows.	Expand LangGraph state machine to **Multi-Agent Swarms (AutoGen/CrewAI)** consensus guardrails.

4. Semester 8 Strategic Roadmap & Timeline (Making It 'Next Level')

Semester 8 will execute a structured 6-phase engineering and research roadmap transitioning Vector-Shield from **TRL 6 (Validated Prototype)** to **TRL 8 (Production-Qualified Enterprise Platform)**:

Phase	Engineering Milestone	Key Technical Deliverables	Timeline & Target
Phase 1	Massive Dataset Scaling	Compile 10,000+ samples across Banking, Healthcare HIPAA, Legal SEC, and Code RAG.	Weeks 1–4 (Dec 2026)
Phase 2	Multi-Modal Vector Shield	Integrate CLIP ViT-L/14 & SigLIP embeddings for visual injection detection in scanned documents.	Weeks 5–8 (Jan 2027)
Phase 3	C++/Rust Envoy Sidecar	Compile PCA-LOF into native Envoy proxy filter (<500µs latency, Kubernetes Helm chart).	Weeks 9–12 (Feb 2027)
Phase 4	White-Box Gradient Defense	Execute GCG and AutoDAN gradient attacks; incorporate adversarial training into ModernBERT.	Weeks 13–16 (Mar 2027)
Phase 5	Vector DB Connectors & EMA	Native connectors for Pinecone, Milvus, Qdrant; online streaming centroid updates (EMA).	Weeks 17–20 (Apr 2027)
Phase 6	Multi-Agent Swarm Defense	Consensus-driven guardrail for AutoGen / CrewAI agent swarms; final thesis & camera-ready paper.	Weeks 21–24 (May 2027)

5. Research Publication Strategy & Intellectual Property

Proposed Paper Title: *'Topographic Anomaly Detection and Stateful Semantic Guardrails in High-Dimensional Embedding Spaces for Securing Autonomous Agentic RAG'*

Target Venues:

- **Tier-1 Cybersecurity:** USENIX Security 2027 / ACM CCS 2027 (Systems & Software Security track).
- **Tier-1 AI / NLP:** ACL 2027 / EMNLP 2027 (Industry & System Demonstrations track).
- **Premier Journals:** IEEE Transactions on Dependable and Secure Computing (TDSC) / IEEE TIFS.

Benchmark Baselines for Paper Acceptance: Llama Guard 3, NeMo Guardrails, SmoothLLM, PromptGuard-86M, Securita.

Intellectual Property & Patents: Indian Provisional Patent Application covering: (1) Non-linear centroid-drift scaled LOF in PCA-reduced embedding manifolds; (2) Cyclic state-machine self-healing for agentic workflows.

6. Formal Supervisor Endorsement Request

AIDS Team 14 respectfully submits this progress report to our project supervisor, **Dr. Lekshmi C. R.** Semester 7 deliverables have met all departmental requirements with high empirical rigor. We request formal approval to proceed with the Semester 8 implementation charter.

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